

CML PROJECT

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0 Introduction

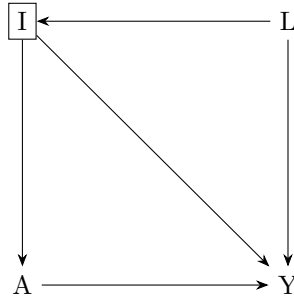
This is our report for the Causal Machine Learning project. For the analyses mentioned below, we used a mixed library of algorithms from the Super Learner package, unless stated otherwise.

As for the distribution of the work, we both looked at all parts; however, in some questions one person clearly took the lead and did most of the work.

Question	Lead
Q1	Shared
Q2	Kylian
Q3	Kylian
Q4	Garai
Q5	Garai

1 Conditional Exchangeability

We start with the hypothesis that, around the time 401(k)'s were introduced, the option to have one was not a variable generally regarded by job seekers, while income obviously was. We also acknowledge that income has an effect on accessibility to the program, but this would have been random otherwise. We will have Y represent total accumulated assets, and L all other variables about the individuals, which we assume account for all other confounding, and we will visualize the assumed relations with a graph:



As we can see, it would have sufficed (at that point in time) to control for income in order to cut the light from A to Y and really measure the causal effect of 401(k) access on total accumulated assets. In other words, we can verify:

$$Y^a \perp\!\!\!\perp A|I$$

2 Estimation of the effect with AIPW, TMLE

When answering this question we presumed it was independent from the one before that.

For estimating the average effect of 401(k) accessibility on net assets (controlling for age, income, family size, years of ed., marriage, dual income, DBP status, IRA participation and home ownership), we believe the average treatment effect is most relevant. It can be useful for policy-making, i.e. for deciding whether broadening 401(k) accessibility is a good idea.

In the future answers we prefer using ATE over ATT. This is because using ATE is more appropriate if we want to answer questions like "what would happen to net assets if eligibility to 401K was expanded". And expanding said eligibility is something a government can do. So we are in that regard trying to answer questions like that from the perspective of a government.

Employing AIPW yielded an estimate of a 7177USD ($CI_{95\%} = [4598, 9755]$) ATE on accumulated assets, that is, having access to a 401(k) would make the average person be about 7 thousand dollars richer. We didn't observe influence functions that were too crazy, the biggest outlier had an influence value of around 40 standard deviations, but there were not many like that either, so we assume the result is appropriate. As for **TMLE**, we had a similar result of 7080USD ($CI_{95\%} = [4480, 9679]$) for the ATE

Employing a naive estimate (generalized linear model) we get the contrasting, blown up result of 19559USD ($CI_{95\%} = [18254, 20864]$), underlining the effect of the confounders.

3 Conditional Covariance

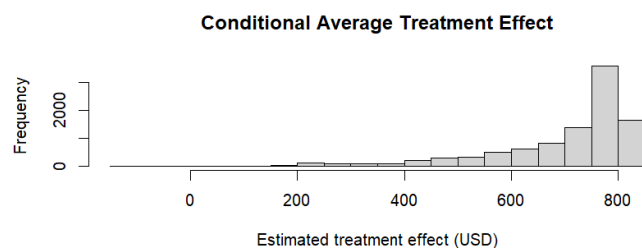
This method of solving the answer was expected to reduce variance and increase power compared to the earlier tests. We might thus expect to see a more significant result now than before. This is our hypothesis test:

$$H_0 : \theta = 0 \quad \text{vs.} \quad H_a : \theta \neq 0$$

The test confirms our expectations, with the following results: The point estimate is $\hat{\theta} = 1680.78$ with standard error $\widehat{SE}(\hat{\theta}) = 264.04$, yielding a z -statistic of $z = 6.37$ and a two-sided p -value of 1.95×10^{-10} .

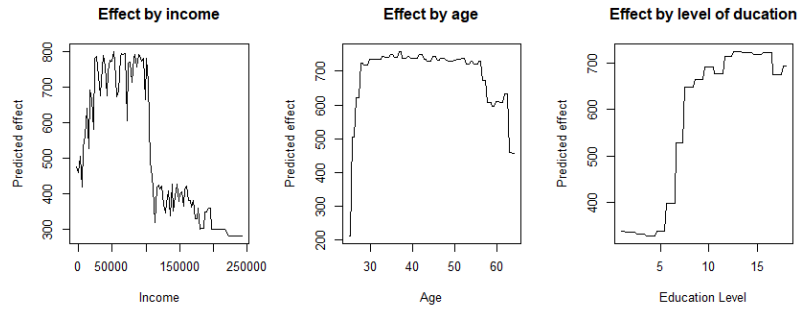
4 CATE with R-learner

This exercise turned out to be specially challenging to us. We started by building the R-learner algorithm "from scratch" but it returned some unsatisfying results, even after making adjustments like clipping the weights etc. Even after switching to using the rlearner library we got results that were centered around 800 USD, with clipped weights and everything double checked. We don't have a conclusion about what went wrong, but we think the results here presented can't be representative, we're still going to carry on. This is what the histogram would look like for the CATE estimation:



We still carried on and modeled the CATE by age, income and level of education, the following is

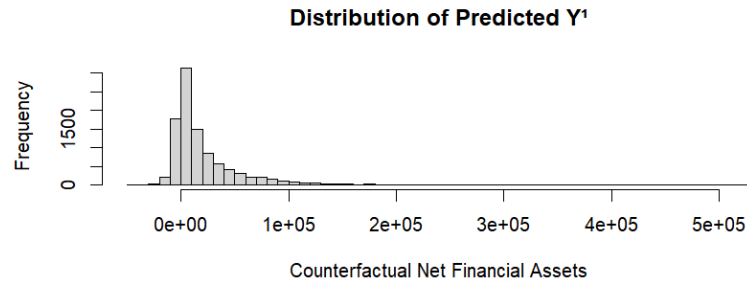
a visualization of the results: We can clearly see that the first graphic indicates a higher treatment



effect among earners that are, so to say, in the middle. Low earners wouldn't be as benefited from 401(k), but as soon as we get into a "comfortable" income bracket the estimated treatment effect seems to blow up, but die down when we reach numbers that could be considered of rich people. As for the second one, it stays pretty similar throughout age (after rising throughout the 20s), but then dies down towards near-retirement ages. The third graph would suggest that, for people with a low education levels the treatment effect would be much milder than for the rest.

5 Counterfactual prediction

First, we made the modified DR-learner "from scratch", clipping $\hat{p}$ values that would have been too low, and got out modified pseudo-outcomes $\tilde{Y}^1$. The estimates came out looking like this(big outlier, but could be reasonable):



In order to simulate the prediction error, we made 500 random samples within our population and looked at the estimated $\tilde{Y}^1$, as well as the $\hat{Y}^1$ that the usual DR-learner would predict, for each sample. Then we calculate the MSE for those values over the samples and came out with an estimation of 84822826 ($CI_{95\%} = [93282, 1381840000]$)

Just like with question four, we deemed this question difficult. We realize the answer is not sensical but we could not find a better answer. This problem was confounded by unstability in the simulations which sometimes led to having to do the same simulation multiple times while every simulation took at least 15 minutes.